

Romain Beauvais

Ph.D. in Acoustics
R&D Engineer - Signal Processing & Data Science

PORTFOLIO

romainbv.github.io

CONTACT

Lille, France
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KEY SKILLS

Physical acoustics & wave propagation
Numerical methods (FDTD, TVD/WENO)
Signal processing
AI / ML - detection & classification
Measurement & testing (PIV, instrumentation)
Multiphysics simulation

TECH STACK

Python (NumPy, pandas, Numba, Pydantic)
C/C++ (pybind11), Matlab, SQLite
COMSOL, SolidWorks, AutoCAD, OnShape
LaTeX · Linux (Ubuntu)

LANGUAGES

French - native
English - fluent (written & spoken)
Spanish - written

CERTIFICATIONS

LASER safety (CERLA)
Electrical certification (H0B0)
Driving licence (B) · BAFA

INTERESTS

Music · Basketball · Climbing

R&D Engineer · Acoustics · Signal Processing · Data Science

PROFILE

Ph.D. in acoustics and R&D engineer, spanning physical modeling (wave propagation, fluid-structure coupling, numerical methods) through to experimental validation and the deployment of embedded AI models. A cross-disciplinary profile covering theory, simulation, testing and industrialization.

EXPERIENCE

R&D Engineer - Signal Processing & Data Science 2022 - 2026
Wavely · Lille

- Acoustic event detection using embedded sensors, integrating AI models (detection & classification).
- Industrial cases: gas leak localization, anomaly detection on rotating machines, drone detection & localization.

Acoustics Lecturer (Sound Design) 2022 - 2023
ISCOM / Studio M · Lille

- 32-hour module: physics of sound, psychoacoustics, room acoustics, electroacoustics.

Postdoctoral Researcher - Turbulence 2020 - 2022
LMFL, Centrale Lille · Villeneuve-d'Ascq

- Turbulent flows in a wind tunnel, LASER measurements (PIV), vortex characterization and comparison with CFD.

R&D Engineer - Numerical Simulation 2019 - 2020
SIM Engineering · Lille

- Nonlinear pulsating gas flows in industrial piping systems (FDTD schemes).

CIFRE Ph.D. - Acoustics & Vibration 2016 - 2019
Le Mans Université - SIM Engineering

- Nonlinear effects and fluid-structure coupling in waveguides, applied to reciprocating compressors.

PERSONAL RESEARCH PROJECT

ReedLab - Physical-modeling sound synthesis of brass
Personal project · 2025 - 2026

Solver computing the **Euler equations** to synthesize the sound of brass instruments from a digital score - a physics-based alternative to MIDI generation. Continuation of my PhD and Edinburgh internship work.

EDUCATION

Ph.D. in Acoustics - Le Mans Université 2019
Research Master's in Acoustics - Le Mans Université 2015
BSc in Engineering Sciences (Acoustics) - Le Mans Université 2013

PUBLICATIONS

7 international publications & communications - *Mechanical Systems and Signal Processing*, *Applied Physics Letters*, Forum Acusticum, ISPIV, INTERNOISE. Full list, DOIs and links on the portfolio.